Proactive Memory: Selective State Intervention for Long-Horizon Agents

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Long-horizon agents must preserve requirements, environmental facts, failed attempts, and unresolved subgoals across an expanding trajectory. We study memory as a selective intervention mechanism rather than a passive archive. A dedicated memory agent maintains structured state and emits a concise reminder only when retained information is likely to affect the next action. This separation keeps the action model unchanged while reducing irrelevant context. Controlled experiments show that selective intervention improves completion rates and avoids the token overhead of exposing the full memory bank at every step. The template demonstrates the native FAIR preprint typography, title card, captions, equations, tables, and references.

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Code: https://example.org/project

1 Introduction

Long-horizon agents increasingly operate over trajectories containing requirements, environmental facts, prior attempts, intermediate diagnoses, and open subgoals. Although these observations may remain in the transcript, they often stop influencing the next action when they matter. Existing memory systems address storage and retrieval (Lewis et al., 2020; Park et al., 2023), but reliable behavior also requires deciding when remembered state should re-enter the control loop.

We study memory as a selective intervention mechanism. A compact memory agent tracks durable state alongside the action agent and emits a targeted reminder only when retained information is likely to change the next decision. The design separates memory maintenance from action generation and remains compatible with unmodified agent harnesses.

Our contributions are threefold. First, we identify behavioral state decay as an access failure rather than a storage failure. Second, we introduce a two-stage architecture that maintains structured state and selects interventions. Third, we evaluate the design under controlled long-horizon tasks and report consistent gains without modifying the action model.

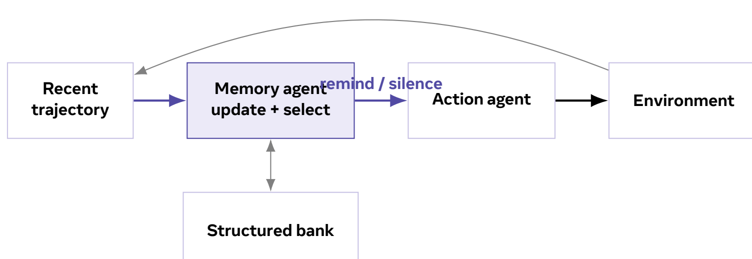

Figure 1 The memory agent maintains structured state and selectively intervenes in the action loop.

2 Related Work

2.1 Long-horizon language agents

Long-horizon evaluations expose failures that are not captured by single-turn reasoning benchmarks. Agents may repeat failed actions, lose task requirements, or retry a near-identical implementation path. Our setting focuses on this form of state decay during extended tool use (Yao et al., 2023).

2.2 Memory and context management

Retrieval-augmented systems improve factual grounding and knowledge access, while agent memory systems organize experience into durable records. These mechanisms decide what to store and retrieve; our formulation additionally asks whether retrieved state should actively modify the next decision.

3 Method

Let H_t denote the recent trajectory and B_t the structured memory bank. The memory agent first produces a sequence of bank edits

$$B_{t+1} = U_\phi(B_t, H_t), \tag{1}$$

then selects an intervention r_t or the null action $\varnothing$:

$$r_t \sim \pi_\phi(\cdot \mid B_{t+1}, H_t), \qquad r_t \in \mathcal{R} \cup \{\varnothing\}. \tag{2}$$

The action agent receives the reminder as transient context. Its base instructions, tools, and decoding procedure remain unchanged.

3.1 Structured memory bank

The bank contains a private status field, stable knowledge entries, and procedural records of attempts and outcomes. Separating these components prevents internal progress tracking from polluting the action agent's context while preserving evidence needed for future decisions.

3.2 Selective intervention

Silence is an explicit action. If no retained state is likely to affect the next decision, the action agent continues without additional context. This decision avoids turning the memory bank into a second, permanently attached prompt.

Theorem 1 (Bounded intervention overhead) *If each reminder contains at most k tokens and the policy intervenes on a fraction ρ of T steps, then total intervention overhead is at most $\rho T k$ tokens.*

4 Evaluation

We compare passive retrieval, full-bank exposure, always-on reminders, and selective intervention. Values below are illustrative and demonstrate the original table treatment.

Table 1 Long-horizon task performance under different memory interfaces.

Variant	Pass@1 ↑	Macro ↑	Tokens ↓
Action agent baseline	49.1	57.5	**1.00**
Passive retrieval	52.6	61.5	1.18
Always-on reminder	58.8	63.5	1.42
Selective intervention	**60.2**	**64.3**	1.11

Selective intervention provides the strongest balance between task success and context overhead. Full-bank exposure preserves information but introduces irrelevant state, while forced reminders remove the policy's ability to remain silent.

5 Discussion

The mechanism is intentionally modular. It can be evaluated with different action models, memory schemas, and intervention budgets. The central design question is not simply whether relevant information can be retrieved, but whether it should become active at the present decision boundary.

6 Conclusion

Memory for long-horizon agents is not only a storage problem. It is also a control problem: retained state must become active at the right moment. A separate memory agent provides a practical way to maintain state and intervene without modifying the action model.

References

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